

AUTOMATA

VIRAL POTENTIAL ANALYZER

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1. Introduction

Many content creators struggle to estimate whether their content has strong engagement potential before publishing. This project develops an analyzer that evaluates content characteristics and provides improvement suggestions.

2. Objective

- Analyze content features affecting engagement
- Calculate a viral potential score
- Use automata to model the evaluation process
- Provide recommendations for improving content performance

3. Dataset

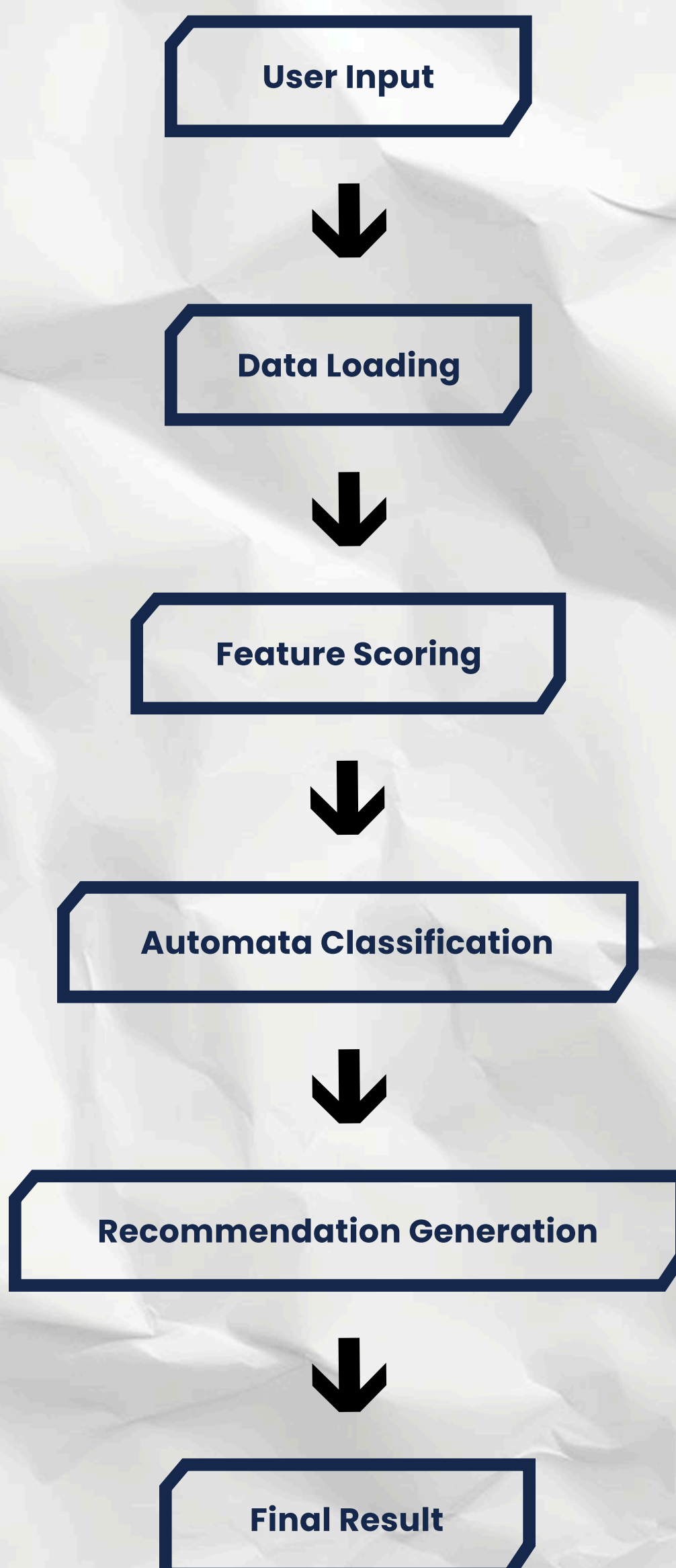
In this project, a custom dataset was manually created because the system requires specific rules and behaviors based on the project requirements. The dataset was designed to support content evaluation, scoring, and recommendation processes.

	Feature	Description
Content	content_type	Type of content
	category	Content category
	points	Score
Hashtag	hashtag	Hashtag name
	trend_level	Popularity
	points	Score
	category	Relevance
Audio	audio_name	Audio title
	trend_level	Popularity
	points	Score
Length	min	Minimum duration
	max	Maximum duration
	points	Score
Posting-Time	start	Hashtag name
	end	Popularity
	points	Score
	engagement	Level of engagement

4. Limitations

- The scoring system is based on predefined datasets rather than live social media data
- Trending hashtags and audio change frequently, so the datasets would need regular updates
- The project predicts viral potential, not guaranteed virality, because many external factors (account size, audience behavior, platform algorithms, etc.) also influence performance

5. Methodology



6. Automata Design



7. Results – Viral

```
=====
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 11

Enter hashtags for your content: fyp, viral,trending,foodie,cooking
```

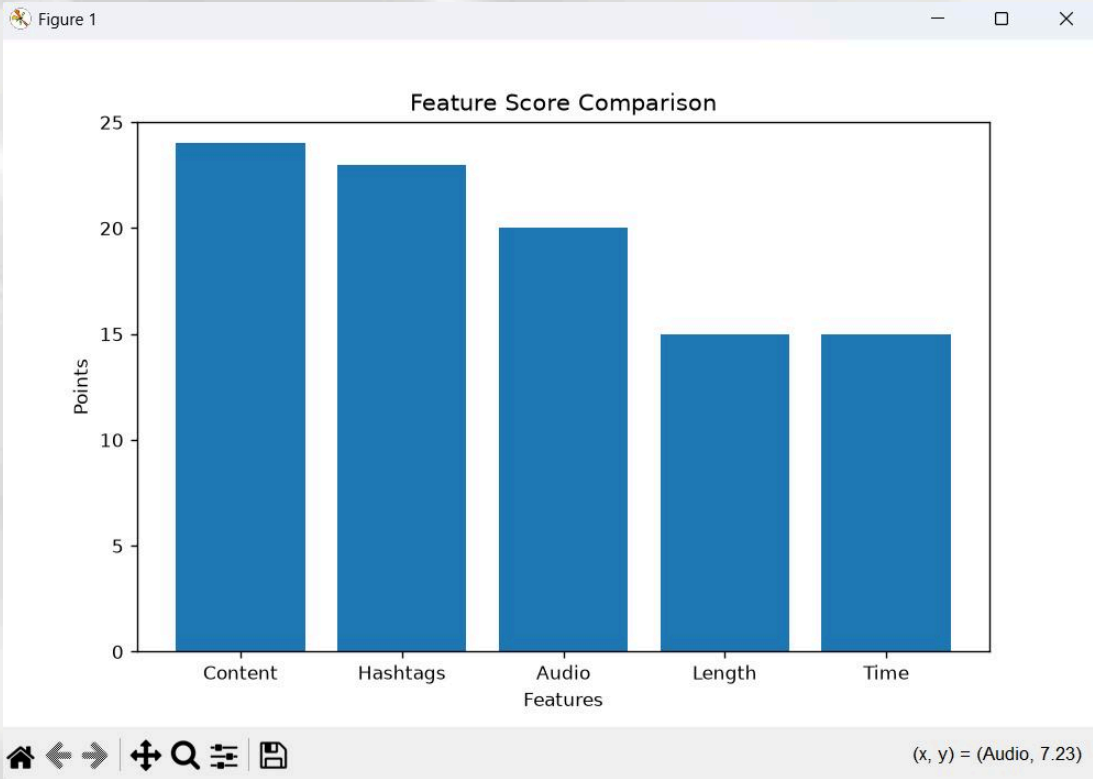
```
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 3

Video Length(seconds): 20

Posting hour (0-23):13

Analyzing...
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```
=====
ANALYSIS RESULT
=====

Prediction   : VIRAL
Total Score  : 97/100

-----
Feature Scores
-----
Content Type : 24
Hashtags     : 23
Audio        : 20
Video Length : 15
Posting Time  : 15

-----
Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
VIRAL
RECOMMENDATION
END

-----
Insights
-----
✓ The selected content type has good engagement potential.
✓ 2 hashtag(s) closely match the selected content.
✓ 3 general hashtag(s) detected.
✓ The selected audio has a good trend popularity.
✓ The video length is within the recommended range.
✓ The selected posting time falls within a peak engagement period.

-----
Recommendations
-----
No recommendations. Great job!
```

7. Results – High

```
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 8

Enter hashtags for your content: python, ai, fyp
```

```
Select Content Type (number): 8

Enter hashtags for your content: python, ai, fyp

Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

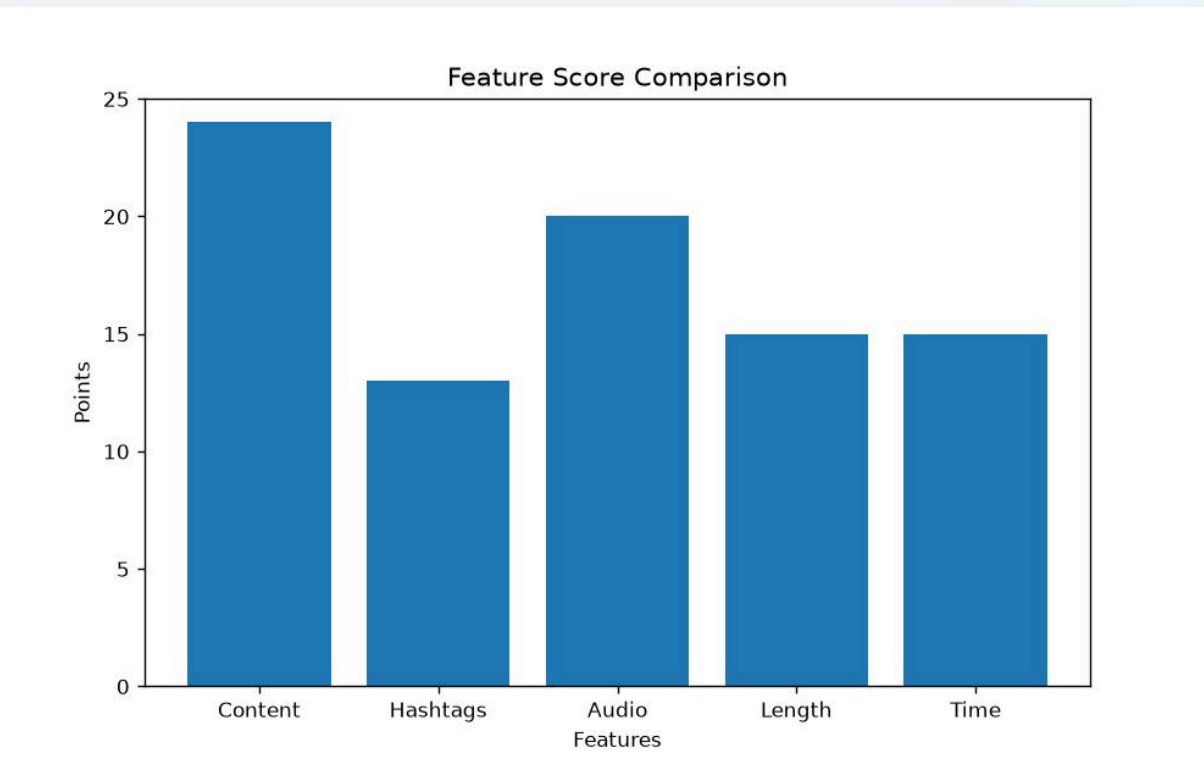
Select Audio (number): 2

Video Length(seconds): 20

Posting hour (0-23):18

Analyzing...
```

Figure 1



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=====
ANALYSIS RESULT
=====

Prediction   : HIGH
Total Score  : 87/100
-----
Feature Scores
-----
Content Type : 24
Hashtags     : 13
Audio        : 20
Video Length : 15
Posting Time : 15
-----
Automata State History
-----
START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
HIGH
RECOMMENDATION
END
-----
Insights
-----
✓ The selected content type has good engagement potential.
✓ 2 hashtag(s) closely match the selected content.
✓ 1 general hashtag(s) detected.
✓ The selected audio has a good trend popularity.
✓ The video length is within the recommended range.
✓ The selected posting time falls within a peak engagement period.
-----
Recommendations
-----
No recommendations. Great job!
```

7. Results – Medium

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=====
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 6

Enter hashtags for your content: books,study
```

```
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 7

Video Length(seconds): 60

Posting hour (0-23):10

Analyzing...
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Recommendations
-----

• Consider using hashtags more closely related to your content.
Suggestions:
- #studytok
- #learnontiktok
- #education
Reason: Some hashtags are less relevant to the selected content category.

• Consider adjusting the video length.
Suggestions:
- 15-30 seconds
Reason: Videos between 15 and 30 seconds receive the highest retention score.

• Consider posting during a peak engagement period.
Suggestions:
- 12:00-14:00
- 17:00-22:00
Reason: Peak posting times receive higher engagement scores in this project.
```



```
=====
ANALYSIS RESULT
=====

Prediction   : MEDIUM
Total Score  : 58/100

-----
Feature Scores
-----

Content Type : 24
Hashtags     : 0
Audio        : 15
Video Length : 9
Posting Time : 10

-----
Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
MEDIUM
RECOMMENDATION
END

-----
Insights
-----

✓ The selected content type has good engagement potential.
✓ 1 hashtag(s) are not found in the dataset.
✓ The selected audio has a good trend popularity.
```

7. Results – Low

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=====
CONTENT VIRAL POTENTIAL ANALYZER
=====

Available Content Types:
1. Comedy
2. Memes
3. Music
4. Dance
5. Pets
6. Education
7. Gaming
8. Technology
9. Beauty
10. Fitness
11. Food
12. Science
13. Art
14. Photography
15. Animation
16. Lifestyle
17. Travel
18. Fashion
19. DIY
20. Productivity
21. Finance
22. Business
23. History
24. Motivation
25. Automotive
26. Books
27. News
28. ASMR

Select Content Type (number): 27

Enter hashtags for your content: gaming, random
```

```
Available Audio Tracks:
1. Sigma Beat
2. Future Bass Drop
3. Dreamcore Loop
4. Speed Up Night
5. Hyper Pop Mix
6. Summer Anthem
7. Lofi Study
8. Soft Piano
9. Happy Pop
10. Night Drive
11. Chill Guitar
12. Acoustic Morning
13. Epic Cinematic
14. Jazz Mood
15. Relax Beats
16. Nature Sounds
17. Rain Ambience
18. Ocean Waves
19. Old Remix
20. Vintage Piano

Select Audio (number): 17

Video Length(seconds): 120

Posting hour (0-23):2

Analyzing...
```

```
-----
Recommendations
-----

• Consider using hashtags more closely related to your content.
Suggestions:
- #news
Reason: Some hashtags are less relevant to the selected content category.

• Consider using a trending audio.
Suggestions:
- Sigma Beat
- Future Bass Drop
- Dreamcore Loop
Reason: Trending audio receives a higher score in the evaluation model.

• Consider adjusting the video length.
Suggestions:
- 15-30 seconds
Reason: Videos between 15 and 30 seconds receive the highest retention score.

• Consider posting during a peak engagement period.
Suggestions:
- 12:00-14:00
- 17:00-22:00
Reason: Peak posting times receive higher engagement scores in this project.
```

```
=====
ANALYSIS RESULT
=====

Prediction   : LOW
Total Score  : 20/100

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Feature Scores
-----

Content Type : 12
Hashtags      : 0
Audio         : 5
Video Length  : 3
Posting Time  : 0

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Automata State History
-----

START
CONTENT_ANALYSIS
HASHTAG_ANALYSIS
AUDIO_ANALYSIS
QUALITY_ANALYSIS
SCORING
CLASSIFICATION
LOW
RECOMMENDATION
END

-----
Insights
-----

✓ The selected content type has moderate engagement potential.
✓ 1 hashtag(s) are not found in the dataset.
✓ The selected audio has moderate trend popularity.
```

